



Digital immortality

Incipient technologies may allow humans to escape the confines of physical bodies, and live in the cloud. <https://digit.in/mar20-59>



The CR society

A community around the calorie restriction diet, with forums, yearly conferences, and research tracking. <https://digit.in/mar20-60>

72 years to around 100 years, or at best 120 years at the present. Bear in mind that there is a major gap between research and practical applications that is suitable for humans. While many experiments have extended the life spans of lab animals by 50 percent, most successfully on mice, none of these treatments have yet translated to human beings. For example, Michael R Rose, a researcher from the University of California, Irvine, conducted a landmark experiment where he progressively increased the lifespan of fruit flies. In the first generation, he picked middle-aged fruit flies that gave birth. In the next generation, he picked fruit flies that gave birth when they were old. In successive generations, Rose kept picking fruit flies that lived longer, and reproduced later in life. While this was successful in fruit flies, such an approach may not be practical in humans.

WE ARE AS OLD AS OUR ARTERIES

One of the telltale signs of ageing is seen in the calcification of arteries, which are bone-like deposits of calcium phosphate crystals along the walls of the arteries. This causes the arteries to harden, and is also responsible for some of the problems associated with old age, such as dementia, high blood pressure, as well as an increased risk of heart diseases and strokes. Calcification also restricts blood flow to all the organs within the body. The smaller blood vessels simply wither and die. This results in low levels of oxygen throughout the body, as well as a buildup of toxins. It is this process that causes the muscles in the elderly to shrink and grow weaker, medically known as sarcopenia. Without say, a birth certificate, measuring the extent of calcification allows for an accurate measurement of a person's age. Even the age at which ancient Egyptian mummies died can be guessed by having a look at the extent of calcification of the arteries. Till recently, the exact process by which the calcification occurs, was not known. Researchers

from King's College of London and Cambridge University published a paper in Cell Reports in June 2019 which outlines the mechanism for calcification of the arteries. As a part of regular DNA repair, cells produce a molecule known as PAR (the full form is a matryoshka doll of acronyms, and is as such irrelevant for the purposes of this article). PAR is released when cells die, and the molecules bind strongly with calcium. As they acquire calcium within the circulatory system, they form into increasingly large droplets, eventually crystallising on the walls of the arteries, a process known as biomineralisation. Now, within bone tissue, this is an essential process to harden the skeleton. However, in the circulatory system, it just causes an increasing number of problems as the body grows older. The researchers then went on to investigate ways in which they could prevent the PAR molecules from crystallising within the arteries. The researchers found that minocycline, an antibiotic commonly used to treat acne, can inhibit the calcification of arteries. This is a promising drug, and the researchers are moving into clinical trials.

REVERSING SARCOPENIA

Another crucial aspect related to ageing that causes the muscles to atrophy and reduces blood flow, is that the body starts growing new blood vessels. After a series of experiments, researchers at Harvard discovered that blood flow within the body begins to reduce after cells stop producing a protein known as sirtuin1, shortened to SIRT1. The reduction in SIRT1 is in turn caused by a reduction of nicotinamide adenine dinucleotide (NAD⁺), a molecule responsible for protein interactions and DNA repair. SIRT1 is known to delay ageing in organisms as different as yeast and mice. NAD⁺ is known to decline with age. Within muscles, SIRT1 is responsible for the growth of capillaries, the smallest blood vessels in the body. As both SIRT1 and NAD⁺ in the body reduce with age, the muscles also deteriorate.

The researchers developed a treatment using nicotinamide mononucleotide (NMN), a derivative of niacin, previously known for its involvement with DNA repair, which is a precursor to NAD⁺. Also included in the treatment regimen was sodium hydrosulfide (NaHS), a precursor to hydrogen sulfide, which also boosts SIRT1 production. The procedure allowed for 32 month old mice (the equivalent of 90 human years) to run for twice as long as untreated mice of the same age. The mice treated with only NMN ran for 1.6 times longer. The treatment in mice at least, proved effective at maintaining capillary growth into old age. However, the researchers warn of the dangers of using treatments that increase blood flow, as the same treatment can encourage tumours to grow faster, by providing them with more nutrition. However, the research is promising in the way that it opens up avenues for future therapies, that will allow the elderly to remain active in their old age.

GO ON A DIET

One of the oldest life extension treatments known to science, is that of calorie restriction. Essentially, it means eating 30 or even 50 percent less than a regular person, but without any compromises on nutrition. If someone does it throughout their lives, they would be ridiculously thin, will always feel cold, and will be perpetually hungry. It's a diet of extreme abstinence, and is also called a fasting diet. Such a diet was known to extend the life spans of mice by 50 percent in an experiment in 1934. A 2017 experiment on rhesus monkeys also showed an extension of the life span. The results have been replicated in worms, microbes, spiders and insects, but not with humans. However, scientists are still in the dark about the exact mechanism by which calorie restriction manages to increase the life spans in almost all the animals that it has been tested on. If this mechanism is understood, scientists may be able to devise a pill that mimics the effects of fasting. 